Supporting Information

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1 Orthogonal LISH mechanism-of-action benchmark

This section provides the complete protocol and results of the phenotype-only LISH benchmark that is summarized in the main text. It is an orthogonal pharmacological reference, not a validation of the molecular representations studied in the main benchmark.

1.1 Data and preparation

We used the public LISH MoA training release as a separate, phenotype-only benchmark; the public challenge repository is available at https://github.com/LISHarvard/moa_challenge (accessed 12 August 2026). The preparation retained 772 gene-expression features, 100 cell-viability features, and 3 experimental-condition summary columns (exposure time, high-dose fraction, treatment-type fraction), and aggregated repeated assay observations to the drug level.

The scored endpoint comprised 206 MoA labels for 3 289 drugs; labels were defined at the drug level from observed MoA-associated annotations. Because the public release used here did not contain a versioned drug–SMILES mapping, the benchmark was not used to train or evaluate molecular fingerprints, GNNs, or transformers. A prespecified sensitivity analysis excluded vehicle-control samples, yielding 3 288 drugs. All fold records were audited for missing or non-finite metrics before interpretation. The endpoint is described as MoA-associated prediction because it does not establish causal target engagement [1].

1.2 Protocol

We fitted an unweighted per-label logistic model independently for each of the 206 labels, evaluated with five random seeds and five drug-grouped folds, and reported mean column-wise log loss as the primary metric with macro-AUPRC, macro-AUROC, mean Brier score, and expected calibration error as secondary metrics.

1.3 Results

With vehicle controls retained, the phenotype-only baseline achieved the metrics reported in Table 1 across 25 fold–seed evaluations. Removing vehicle controls changed macro-AUROC by only -0.0018, and the other metrics moved by less than 0.002 in absolute value, indicating that the phenotype-driven baseline is robust to control exclusion.

1.4 Interpretation boundary

These metrics summarize the phenotype-only reference; they are **not numerically comparable** with the molecular ROC-AUC results of the main manuscript because the task, labels, features, aggregation unit, and primary metric differ. The labels indicate MoA-associated prediction rather than causal target engagement. The LISH analysis therefore serves as an orthogonal pharmacological reference and control-sensitivity analysis, not as a molecular-model validation or evidence of causal mechanism.

Table 1: Phenotype-only LISH baseline metrics, mean across 25 fold–seed evaluations (five seeds × five drug-grouped folds).

Metric	With controls	Without controls	Δ
Mean column-wise log loss	0.023 78	0.023 89	+0.000 11
Macro-AUROC	0.643 5	0.641 7	−0.001 77
Macro-AUPRC	0.142 8	0.141 2	−0.001 52
Mean Brier score	0.003 74	0.003 75	+0.000 01
Expected calibration error	0.003 76	0.003 81	+0.000 05

2 Post-hoc calibration metrics for the molecular benchmark

Calibration metrics were computed post-hoc from the archived fold-level test predictions of the extended campaign (five seeds × five folds per configuration; 99 180 pooled test records per configuration). No model was retrained; the estimates therefore describe the calibration of the deployed checkpoints, pooled over seeds and folds, and are reported as a pooled-calibration estimand. Expected calibration error (ECE) and maximum calibration error (MCE) use 10 equal-width bins; the Brier score is the mean squared error between predicted probability and label.

Table 2: Post-hoc calibration metrics (pooled over 25 fold–seed evaluations per configuration). ECE and MCE rise from the random split to the scaffold and novel-partition splits for every model family, indicating that confidence degrades under scaffold shift even when ranking (ROC-AUC) is only moderately affected.

Split	Model	ECE	MCE	Brier
Canonical random	GIN	0.029 0	0.069 9	0.102 55
	GIN-TFP (native)	0.034 4	0.071 6	0.105 24
	GIN-TNE (native)	0.036 7	0.093 6	0.112 86
	ChemBERTa	0.057 8	0.144 0	0.104 66
Canonical scaffold	GIN	0.065 4	0.162 3	0.148 59
	GIN-TFP (native)	0.074 1	0.175 4	0.147 69
	GIN-TNE (native)	0.080 7	0.194 6	0.151 08
	ChemBERTa	0.122 5	0.254 6	0.169 09
Novel partitions (range)	GIN	0.0777–0.0888	0.1814–0.1854	0.15359–0.15919
	GIN-TFP (native)	0.0741–0.0895	0.1737–0.1890	0.14922–0.15564
	GIN-TNE (native)	0.0798–0.0820	0.1813–0.2078	0.15305–0.15539
	ChemBERTa	0.1172–0.1320	0.2401–0.2715	0.16699–0.17508

The degradation is consistent across families: ECE roughly doubles from the random split to the scaffold split and widens further on the novel partitions, and ChemBERTa — the arm with the largest ROC-AUC fold-level spread — also shows the largest calibration error under scaffold shift. These estimates are secondary robustness evidence; they do not alter the canonical ROC-AUC results and are not presented as calibrated probabilities for prospective use. Per-configuration reliability summaries are provided in the released data package.

3 Light GNN hyperparameter sensitivity on the scaffold split

To probe whether the canonical negative result depends on the specific GIN capacity choice, we retrained the base GIN on the frozen canonical scaffold split with two configurations bracketing the canonical setting (hidden dimension 128, dropout 0.1): a smaller model (hidden 64, dropout

0.2) and a larger model (hidden 256, dropout 0.1). The protocol was otherwise identical to the canonical benchmark (five seeds $\times$ five folds, same optimizer constants, epoch budget, and early-stopping rule). These sensitivity runs are secondary analyses and leave all canonical estimates unchanged.

Table 3: Base-GIN scaffold-split sensitivity to hidden dimension and dropout (secondary analysis; 25 fold–seed records per configuration). The canonical configuration (hidden 128, dropout 0.1) reached ROC AUC 0.804 7.

Configuration	ROC AUC (mean of 25)	Per-seed SD	Fold-level SD
Hidden 64, dropout 0.2	0.808 1	0.011 8	0.038 6
Hidden 128, dropout 0.1 (canonical)	0.804 7	0.014 1	0.039 5
Hidden 256, dropout 0.1	0.800 0	0.016 0	0.037 0

All three configurations remain clearly below the ECFP4–RF scaffold reference (ROC AUC 0.830 0), and the spread across configurations (0.800 0–0.808 1) is small relative to the gap to the fingerprint reference (0.022–0.030). Doubling or halving the hidden capacity, or changing the dropout rate, therefore does not close the scaffold-split gap on this panel. This sensitivity check bounds, but does not eliminate, the possibility that some untested architecture or budget would perform differently; it supports reading the canonical negative result as a property of the representation comparison rather than of one specific capacity choice.

4 Paired differences against the ECFP4–RF reference

The main text reports the paired model-minus-reference differences with their permutation-based intervals. For completeness, Table 4 archives the audited paired seed-level contrasts with 95% intervals derived from the paired t statistics (5 per-seed means per arm). All eight contrasts are negative and exclude zero at the 95% level after Benjamini–Hochberg correction.

Table 4: Paired model-minus-ECFP4–RF AUC differences with 95% intervals (secondary table; derived from the audited paired seed-level statistics).

Split	Model	ΔAUC	95% interval	Adjusted p
Random	GIN	−0.033 5	−0.036 8–−0.030 2	9.9×10^{-6}
Random	GIN–TFP	−0.034 9	−0.036 5–−0.033 3	6.0×10^{-7}
Random	GIN–TNE	−0.051 5	−0.053 9–−0.049 1	6.0×10^{-7}
Random	ChemBERTa	−0.031 2	−0.032 5–−0.029 9	6.0×10^{-7}
Scaffold	GIN	−0.025 3	−0.043 4–−0.007 2	0.033 0
Scaffold	GIN–TFP	−0.016 2	−0.029 0–−0.003 4	0.033 0
Scaffold	GIN–TNE	−0.021 0	−0.040 1–−0.001 9	0.037 8
Scaffold	ChemBERTa	−0.043 3	−0.049 9–−0.036 7	0.000 2

These intervals are consistent with the permutation analysis of the main text. They are descriptive summaries of the evaluated seed–fold grid and do not extend to other panels, splits, or architectures.

References

[1] Maria-Anna Trapotsi, Layla Hosseini-Gerami, and Andreas Bender. Computational analyses of mechanism of action (moa): data, methods and integration. *RSC Chemical Biology*, 3(2): 170–200, 2022. doi: 10.1039/D1CB00069A.